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REVIEW

CHEMISTRY & BIODIVERSITY

Risk Associated with the Use of Selected Ingredients in Food Supplements

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This review focuses on four new product categories of food supplements: pre-workout, fat burner/thermogenic, brain/cognitive booster, and hormone/testosterone booster. Many food supplements have been shown to be contaminated with unauthorized substances. In some cases, the ingredients in the new categories of dietary supplements were medicinal products or new synthetic compounds added without performing clinical trials. Some of the new ingredients in dietary supplements are plant materials that are registered in the pharmacopoeia as herbal medicines. In other cases, dietary supplements may contain plant materials that have no history of human use and are often used as materials to 'camouflage' stimulants. In the European Union, new ingredients of dietary supplements, according to European Food Safety Authority or unauthorized novel food. Furthermore, selected ingredients in dietary supplements may be prohibited in sports and are recognized as doping agents by World Anti-Doping Agency.

Keywords: doping, food supplements, new components, SARM, steroids, stimulants, toxicology.

1. Introduction

Preparations that can be used to supplement the human diet with vitamins, minerals, or nutrients such as proteins, carbohydrates, and fats were used in the first half of the 20th century in the United States and Europe, and the main consumers were pregnant or breastfeeding women, infants and growing children, and older people.[1,2] Since the 1940s, supplement the human diet have been used as ergogenic support in athletes.[3,4]

The generally accepted date on which dietary supplement market was officially started in the United States is October 25, 1994, following the approval of 'The Dietary Supplement Health and Education Act of 1994 (DSHEA)'.[5] The approval of the act 'Regulation (EC) No. 258/97 of the European Parliament and of the Council on 27 January 1997 concerning novel foods and novel food ingredients' started the development of the dietary supplement market in Europe.[6]

Since then, the dietary supplement market has witnessed a considerable growth in its monetary value, and many new products have been introduced. The advertisement of dietary supplements on radio, print, TV, and the Internet contributed to their increased sales, both offline (pharmacies, drug stores, gas stations, and grocery stores) and online (websites).[7,8] Several new categories of dietary supplement products were introduced, including pre-workout, fat burner/thermogenic, brain/cognitive booster, and hormone/testosterone booster.[9,10]

Recommendations for diet supplementation for physically active people are specified by the International Society of Sports Nutrition (ISSN), the International Olympic Committee (IOC), and the Australian Institute of Sport (AIS).[11,12] Some of the new ingredients used in food supplements were contrary to the ones determined by the regulations of pharmaceutical law. In some cases, new ingredients in dietary supplements were actually doping agents and violated the regulations of the International Olympic Committee (IOC) and later, the World Anti-Doping Agency (WADA).[13–15]

Authorized ingredients in dietary supplements in European Union (EU) countries are determined by the European Food Safety Authority (EFSA). The ingredients can be accessed in the EFSA novel food catalogue or Scientific Report – Compendium of

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Botanicals, which is a database of botanical substances of possible concern for human health. Furthermore, governmental organizations in EU countries have approved certain recommendations, reports, and opinions regarding the safety of active ingredients contained in dietary supplements, e.g., Report of the Scientific Committee of the Spanish Agency for Food Safety and Nutrition (AESAN); Opinion of the Panel on Nutrition, Dietetic Products, Novel Food and Allergy of the Norwegian Scientific Committee for Food Safety; Opinion of the French Agency for Food Environmental and Occupational Health and Safety; Ministerial Decree, The Kingdom of Belgium; Comments of the German Federal Institute for Risk Assessment (BfR); and the Chief Sanitary Inspectorate, Poland. The Rapid Alert System for Food and Feed (RASFF) also plays an important role in controlling unauthorized ingredients in food, including dietary supplements, in the EU. In Poland, in 2017, the Supreme Audit Office published a report on the quality and safety of dietary supplements.[16–18]

The present review focuses on unauthorized and authorized natural ingredients and active substances isolated from them and synthetic/semi-synthetic ingredients found in dietary supplements in Europe, which have been linked to side effects and legality issues. Because online shopping for dietary supplements is widely available, dietary supplements available in the United States, Australia, and selected Asian countries were also considered. The problem of unambiguous determination of whether the questioned ingredient in dietary supplements is of natural or synthetic origin was raised. The present review provides statistical data in relation to the prevalence of dietary supplement consumption and the frequency of use of dietary supplements by the general population and specific subpopulations, i.e., professional, and amateur athletes (in various age categories), professional soldiers, and students. The risk related to the use of dietary

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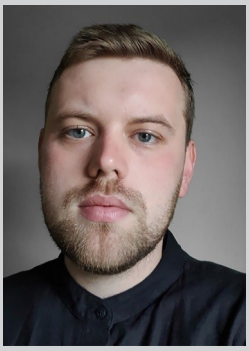
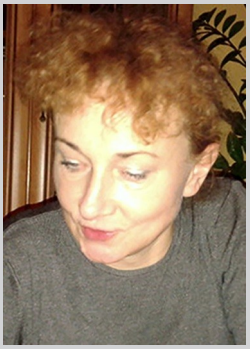


Jan Lazur, student of the 5th year of Pharmacy in Jagiellonian University Medical College, Faculty of Pharmacy. His scientific career has started from the first year of Pharmacy under supervision of Prof. Bożena Muszyńska. President of scientific study group of medicinal plant and mushroom biotechnology. His research interests include applied phycology: prediction of drugs biodegradation from aquatic environment using in vitro methods and analysis of secondary metabolites from dietary supplements containing microalgae.



Bożena Muszyńska, professor of medical science at the Faculty of Pharmacy of the Jagiellonian University Medical College. She is a pharmacist, mycologist, and botanist. The member of Polish Mycological Society and member of the Board of the Medical Mycology Section. From last 30 years the most important aim of her research is to prove the therapeutic and dietary value of mushrooms (Basidiomycota), while the aim of research in the field of mushroom biotechnology is to propose mycelial cultures as a rich, alternative source for bioactive components obtaining. Another subject of research was an effect of edible mushrooms fruiting bodies processing on the content of selected biologically active compounds and their supplementation by mycelia in active substances in order to obtain a material with better dietetic and medicinal properties. An important direction of her research in the field of mycochemistry is the assessment of biological activity (inter alia: anti-inflammatory, antioxidant, antidepressive, immunostimulatory) of extracts obtained from the fruiting bodies and mycelium of the examined species. Other interests were also focused on the evaluation of mycelium from in vitro cultures in terms of bioremediation of the natural environment from xenobiotics by selected mushroom species. She authored more than 150 articles in peer-reviewed international journals, 4 books, and 14 chapters in textbooks and many invited conference (more than 300).

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supplements was considered in the following three areas. The first area discusses the safety of dietary supplement use on the basis of reported adverse effects (including toxic effects). The second area describes the legal issues in the light of the applicable standards in the EU. The third area focuses on the legality (or doping) issue by considering WADA regulations. The article provides a general classification of ingredients prohibited and permitted by WADA regulations. Ingredients that have been not characterized and not included in the WADA Prohibited List are not always safe for health and may conflict with the EU food law. Hence, an ordination has been prepared in line with the regulations of EFSA for authorized and unauthorized ingredients.

Dietary supplements are commonly used in society by people of various age categories – young people, middle-aged people (young adults), and elderly. Studies have been conducted in specific subpopulations using dietary supplements, i.e., professional athletes, amateur athletes, professional soldiers, and students.[19,20]

During the Sydney Olympics in 2000, it was reported that 72% of the athletes were using dietary supplements.[21] In another study, among the 203 students participating in physically active recreation programs, 89% used dietary supplements.[22] Dietary supplements are often used by physically active people, including elite professional athletes and people involved in recreational sports (amateur), in various age groups and in various disciplines.[23,24] A meta-analysis conducted in 2016 showed that elite athletes used dietary supplements more often than amateur athletes.[25]

In Germany, 80% of 228 young professional athletes used dietary supplement.[26] In Poland, 98% of the surveyed adults reported the use of dietary supplements. Women significantly more often used dietary supplements than men. Women also used fat burner dietary supplements more often than men. Advertisement was the main source of knowledge of dietary supplements for the respondents.[27] Among recreationally active people aged 21–31 years, all used dietary supplements. For the majority of respondents, the primary motive for the use of supplements was to improve performance in sports training. In the surveyed group, 84% of the respondents used energy drinks and 60% used supplements as fat burners.[28] Among men aged 18–34 years who trained in competitive martial arts, 66% used dietary supplements, of which 24% of the respondents used caffeine-stimulating dietary supplements, while 8% used fat burners.[29] Among men undergoing strength training for recreation sports, 79% declared that they used dietary supplements. One of the reasons for using dietary supplements is to improve muscle strength and mass.[30] A research study conducted in 2019 showed that in a group of 200 professional special forces soldiers from Poland, 101 participants (50.5%) used dietary supplements. The decision to use supplements was independently made by 79% soldiers, and only 6% soldiers used dietary supplements following the recommendation of professional medical personnel.[31] Among 546 students at Krakow universities, 35% used dietary supplements containing high doses of caffeine, mainly to reduce fatigue, shorten sleep time, and improve concentration during the examination session. The main side effects of consumption of energy boosters by students were insomnia, excessive psychomotor agitation, and muscle contraction and tremble.[32]

2. Ingredients Permitted (or Not Specified) by WADA

2.1. *Natural Ingredients and Isolated Active Substances*

Corynanthe yohimbe *(Yohimbine)*

Yohimbine is an indole alkaloid found in *Corynanthe yohimbe* bark. Native Africans used *C. yohimbe* to treat impotence (erection support). Yohimbine is an α_2-adrenergic receptor antagonist and increases the release of norepinephrine in both the central and peripheral nervous systems. Consequently, increased blood pressure, increased heart rate, and expansion of blood vessels are observed in the peripheral parts of the body. In the 1980s and 1990s, yohimbine was studied as a potential erection enhancer. Morales et al. described the improvement of erections in men suffering from impotence.[33,34] A previous study reported erection improvement in more than 30% of patients.[34] Information on the use of yohimbine and its possible side effects was described in the scientific literature as early as the 1980s. Another study reported cases of submaniacal states following yohimbine intake.[35] Yohimbine has also been described as a *new street drug*. The study reported a case of a young woman who, after consuming yohimbine, experienced dissociation (personality breakdown), paresthesia, and coordination disturbance.[36] A meta-analysis in 1998 concluded that reports on the efficiency and effectiveness of yohimbine and its safety were inconsistent.[37] In the work of Teloken et al. the authors proved the lack of effectiveness of yohimbine in

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the treatment of erectile dysfunction. Attention was given to common adverse effects, including headaches, tachycardia, and anxiety attacks.[38] On the basis of scientific publications, yohimbine generated interest of athletes and physically active people for use as an energy booster (stimulating) agent for reducing fatigue and intensifying fat burning. A previous study showed that in a group of football players who used yohimbine at the dose of 20 mg per day, there was no improvement in performance capacity. A slight effect of this compound on the reduction of adipose tissue was demonstrated.[39] Sommer et al. showed that yohimbine used by healthy volunteers (both physically active and inactive) can have negative effects on the body, as it increases cortisol levels and causes anxiety.[40] The first reports of the presence of yohimbine in dietary supplements were published in 1995. At the same time, yohimbine hydrochloride was used as a drug in the treatment of erectile dysfunction. Yohimbine hydrochloride is a medicinal raw material recommended by the European Pharmacopoeia 2011.[41] Yohimbine in combination with caffeine and ginseng in energy drinks may cause numerous adverse effects.[42] On the basis of a qualitative and quantitative analysis of the ingredients of dietary supplements with their declared content of *C. yohimbe* and yohimbine, a previous study showed that in the analyzed products, the amount of yohimbine was overestimated and was often of synthetic origin (yohimbine hydrochloride).[43] Another study reported neurotoxic effects (convulsions) after yohimbine intake. The most common side effects caused by using *C. yohimbe* or yohimbine include increased blood pressure, tachycardia, anxiety, restlessness, and irritability.[44,45] Yohimbine may interfere with caffeine, antihypertensive drugs, antidepressants, and sympathomimetic drugs, e.g., ephedrine and synephrine. In the EU, *C. yohimbe* or yohimbine has not yet been approved as a component of dietary supplements.[46] In the RASFF panel, yohimbine has been notified as an unauthorized substance in dietary supplements. Yohimbe bark extract is designated as an unauthorized novel food ingredient.[17]

Rauwolfia *sp.* (*Reserpine and its Derivatives*)

Rauwolfia vomitoria and *Rauwolfia serpentina* plant raw materials are used in ethnic medicine of Asian and African countries as a blood pressure lowering agent and a mood stabilizer. Reserpine, one of the first antihypertensive drugs, is obtained from the root/rhizome of *R. serpentina*.[47,48] The *Rauwolfia* species contain indole alkaloids, including reserpine, rauwolscine, yohimbine, and ajmaline. Reserpine is a compound that blocks the postsynaptic transporter for monoamines, namely serotonin, dopamine, and noradrenaline, leading to the inhibition of their reuptake from synaptic clefts. As a result, there is a nonselective increase in monoamine concentration. Reserpine has been traditionally used as a drug for treating hypertension, but because of its several side effects, i.e., nausea, vomiting, bradycardia, depressive or manic states, and impaired coordination, it has been withdrawn and replaced with newer preparations. Ajmaline has also been used traditionally in cardiology as an antiarrhythmic drug. Ajmaline inhibits sodium channels and extends the duration of action potential. Because of its narrow safety margin that leads to the arrhythmogenic effect, it was withdrawn from pharmacotherapy.[47,48] Apart from reserpine, other alkaloids such as ajmaline and yohimbine were detected in the tested food supplements containing *Rauwolfia* sp.[49,50] Another study indicated the potential interactions of dietary supplements containing *Rauwolfia* sp. with antihypertensive and antidepressant drugs (MAOI and SSRI). Alkaloids from *Rauwolfia* sp. may also interact with other components of dietary supplements, e.g., St. John's wort (*Hypericum perforatum*).[51] Reports from Germany describe a case of Parkinsonism and depression in a man who was taking a dietary supplement containing reserpine.[52] *Rauwolfia* sp. are also found in the dietary supplement categories of pre-workout and fat burners. In the RASFF panel and the EU novel food catalogue, *Rauwolfia* sp. is designated as an unauthorized novel food ingredient. In 2020, the RASFF panel registered a large number of reports of dietary supplements containing another species of *Rauwolfia*, namely *Rauwolfia canescens*.[17]

Vinca minor (*Vincamine and its Derivatives*)

Scientific studies have shown that *Vinca minor* (lesser periwinkle) may be present in food supplements.[53–55] *V. minor* is a raw plant material that serves as a source of vincamine. Vincamine is an indole alkaloid that has a vasodilating effect on cerebral vessels and improves blood flow. Vincamine blocks sodium channels and functions as a calcium channel antagonist, i.e., it limits the accumulation of calcium ions within neurons. Vincamine inhibits adenosine reuptake, functions as an inhibitor of the enzyme phosphodiesterase I (PDE I), and increases the concentration of cAMP. Currently, a semi-synthetic derivative of vincamine, namely vinpocetine, is used in pharmacotherapy. Vinpocetine is used as an ingredient in prescription medication

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indicated for the treatment of cerebral insufficiency with people suffering from dementia or after a stroke. The use of vinpocetine is associated with minor adverse effects, including insomnia, hot flushes, dizziness, and decreased blood pressure.[56–58] In 12 healthy volunteers administered with vinpocetine, a shortened reaction time in the performed tasks was observed.[59] The presence of both vincamine and vinpocetine has been shown in the dietary supplements of the categories brain/cognitive booster, pre-workout, and fat burner. Vinpocetine was determined in the amount of 5–10 mg in a single portion in the analyzed food supplements. This dose was the same or greater than that found in prescribed medications. Vincamine and vinpocetine interfere with both anti-hypertensive and anticoagulant drugs.[53–55]

Both vincamine and vinpocetine have been notified as unauthorized substances in food supplements in the RASFF panel.[17] *Vinca* spp., as a whole plant, is mentioned in EFSA Compendium of Botanicals as a botanical substance of possible concern for human health.[60]

Huperzia serrata (*Huperzine and its Derivatives*)

Huperzia serrata (Toothed Clubmoss, *Lycopodium serratum*) has been used in Traditional Chinese Medicine to reduce the effects of fatigue and to improve memory. Historical records show that some side effects of *H. serrata* are also known, in particular vomiting effects and psychomotor coordination disturbance. *Huperzia* (though not often) was also used as a poisonous substance and is relatively complicated to identify. *H. serrata* contains the quinolizidine alkaloid – huperzine A (*Figure 1*). Huperzine is a reversible and selective inhibitor of acetylcholinesterase (AChEI). Huperzine shows affinity for the NMDA receptor for which it acts as an antagonist. Huperzine has also attracted the interest of the army as an alternative antidote in combat gas poisoning by organophosphorus compounds.[61–63]

Orally administered huperzine rapidly crosses the blood-brain barrier and acts on the central nervous system (CNS). Pharmacological activity of huperzine has been observed at microgram level doses, which differentiates huperzine from other reported unauthorized substances in dietary supplements whose doses are in milligrams. This indicates that huperzine has a high potential of biological activity even at low doses. In human studies, huperzine has been administered in doses of 200–500 μg per day. In patients with Alzheimer's disease, huperzine has been shown to improve cognition and alleviate memory deficit.[64–66]

Other substances showing cholinergic activity, such as galantamine or lobeline, were also found in dietary supplements, but in negligible cases. James-Yi et al. confirmed the probability of interactions resulting from the presence of AChEI in dietary supplements, including huperzine and galantamine.[51] Lobeline is a piperidine alkaloid found in the leaves and seeds of *Lobelia inflata*. Lobeline was used as an analeptic agent for resuscitation of electrocuted or drowned persons, because of its stimulatory effect on the respiratory center in the brain. As it has a narrow therapeutic index, it was withdrawn from pharmacotherapy. However, lobeline is present in some food supplements recommended for use in preparations helping to quit smoking and in dietary supplements

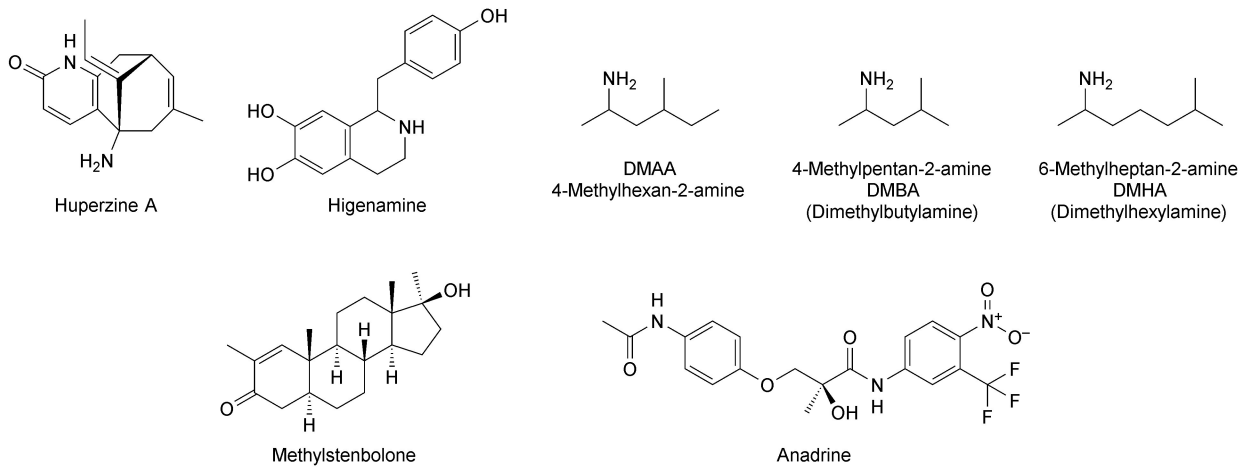

Figure 1. Natural and synthetic ingredients in food supplements for athletes.

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for athletes. A previous study reported the use of lobeline as an agent to enhance the performance of athletes.[67–69]

H. serrata or huperzine has been found in dietary supplements of the categories pre-workout and brain/cognitive booster, and less often in fat burner. Higher doses of huperzine or other AChEI can cause poisoning due to an increase in acetylcholine levels and lead to the so-called cholinergic syndrome that manifests as symptoms of the central and peripheral nervous system, such as vomiting, deliria, impaired psychomotor coordination, and muscle weakening, including respiratory muscles.[63,70,71]

In the RASFF panel, *H. serrata* has been notified as a novel food ingredient. Huperzine is recognized as an unauthorized substance.[17] *H. serrata* is included in the EFSA Compendium of Botanicals as a botanical substance of possible concern for human health.[60]

Piper methysticum (*Kavalactones*)

Methystine pepper (*Piper methysticum*), commonly known as Kava-Kava, is a plant found in the Pacific islands and Australia. *P. methysticum* was used by the people of these regions to make a beverage that has calming, sleeping, and mood regulating properties. The indigenous people also used Kava-Kava as a euphoric agent, which, based on its effects on the human body, is often compared to the effects caused by the use of ethyl alcohol. The raw material is root/rhizome of *P. methysticum*, which contains compounds from the group of kavalactones or kavapyrones: kavain, methisticin, iagonine, and their hydrogenated derivatives.[72,73] Iagonine increases the level of serotonin in synaptic cleft. Methisticin can activate GABA-A and GABA-B receptors. Kavain and iagonine reduce dopamine levels, while their hydrogenated derivatives show an opposite effect, i.e., an increase in dopamine levels.[72–75]

The anxiolytic and antidepressant effects of *P. methysticum* have been observed in the human body after using 250–300 mg of the extract standardized for kavalactones.[76–78] Sarris et al. showed a reduction in anxiety and symptoms of depression in adults using Kava-Kava, which provided 250 mg of kavalactones per day to the body.[79] It has been proven that acute consumption of 300 mg of *P. methysticum* extract (90 mg of kavalactones) in healthy volunteers improves cognitive functions: speed and accuracy in the performed memory tasks and shortening of reaction time.[80]

Kavalactones affect the activity of liver enzymes, which increases the risk of interactions with the drug. Hepatotoxic effects of Kava-Kava have been reported, which are directly related to the raw material parts used, including mainly aerial parts (leaves and bark) of *P. methysticum*.[81–83] The consumption of kava-kava has been shown to impair the ability to drive vehicles.[84]

P. methysticum is listed in the WHO, EMA, and ESCOP monographs. The rhizome of *P. methysticum* has been included in the British and American Pharmacopoeia. In Germany, the Federal Institute for Drugs and Medical Devices (BfArM) has banned the registration of herbal medicinal products containing Kava-Kava several times. In Poland, since 2005, Kava-Kava has been listed as a narcotic substance, and its possession or distribution is subject to criminal law. In 2018, *P. methysticum* was removed from this list of narcotic substances.[72,75]

Dietary supplements containing Kava-Kava from the category of brain/cognitive booster have been identified. It has been speculated that Kava-Kava may be helpful to athletes as a stress-relieving agent, which helps to induce sleep and thus indirectly contributes to overall body regeneration. In some sports such as archery, Kava-Kava has been suggested to help manage emotions and stress, reduce hand tremors, and improve precision, and to exert a positive effect on concentration/focus.[85] Kava-Kava is included in the EFSA Compendium of Botanicals as a botanical substance of possible concern for human health and is recognized as an unauthorized ingredient in dietary supplements in the EU. In the RASFF panel, there is one notification from 2020, namely a dietary supplement containing an unauthorized novel food ingredient – *P. methysticum*.[17,60]

Sceletium tortuosum

Sceletium tortuosum is an African plant whose herb has been used for centuries as a natural medicine for calming and relief from stress and anxiety (in the form of chewed raw material). *S. tortuosum* is more often associated with the name of Kanna. At higher doses, Kanna shows euphoric and psychotropic effects. The active ingredients present in the plant are alkaloids, indole, and pyrrolidine derivatives: mesembrine, mesembrenone, and mesembrenol. The mechanism of the biological activity of the alkaloids present in *S. tortuosum* is based on the inhibition of serotonin reuptake from synaptic clefts, which affects mood and anxiety; this effect is mainly due to mesembrine.

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Mesembrenone showed the greatest affinity as an inhibitor of the enzyme PDE IV. The psychotropic effect is due to the substances extracted into saliva by chewing a dried plant (Kanna) or a fermented plant (Kougoed).[72,86]

Acute oral consumption of 25 mg of an *S. tortuosum* extract is reported to affect the amygdala in humans and reduce anxiety.[87] Supplementation of *S. tortuosum* extract (0.4% of total alkaloids) for 3 weeks led to an increase in executive functions.[88] A single dose of *S. tortuosum* was found to relieve stress/anxiety in healthy volunteers.[89] In a group of healthy, young adults (aged between 20 and 35 years), it was proved that 8-day use of 25 mg/day *S. tortuosum* extract yielded ergogenic benefits in the cognitive domain.[90] *S. tortuosum* extract at the doses of 8 and 25 mg/day was well tolerated in healthy people for 3 months.[91]

Kanna, however, is detected in forensics to be a component of 'designer drugs' or as adulterated (fake) marijuana, which confirms the psychotropic potential of *S. tortuosum*.[92,93]

S. tortuosum is a component of dietary supplements of the pre-workout and brain/cognitive booster categories. *S. tortuosum* is recognized as an unauthorized novel food in the EU and is listed in the EFSA Compendium of Botanicals as a botanical substance of possible concern for human health.[17,60]

Griffonia simplicifolia (*5-HTP*)

Griffonia simplicifolia (African black bean) is a plant whose seeds contain up to 20% 5-hydroxytryptophan (5-HTP). In African countries, *G. simplicifolia* has been used as a tonic and antistress agent in traditional medicine to enhance sexual drive and suppress appetite.[72,94] Rondanelli et al. reported the effect of using *G. simplicifolia* on satiety in overweight women.[72,95,96]

Many studies have been conducted on the active ingredient of *G. simplicifolia*, namely 5-HTP, which is a precursor of serotonin. In the 1960s and 1970s, several studies showed the possibility of using 5-HTP for treating depression. 5-HTP was shown to have an antidepressant effect, which was comparable to that of antidepressant drugs. Current studies, however, have shown that long-term use of 5-HTP may contribute to the washout of catecholamines, namely noradrenaline and dopamine, stored in neurons, and the imbalance in the monoamine system causes a predisposition to the development or exacerbation of existing diseases related to the catecholamine system, such as depression, anxiety, or affective disorders.[97,98] In a group of overweight women, the use of 5-HTP was found to reduce food consumption and promote weight loss.[99] In overweight patients with noninsulin-dependent diabetes, 5-HTP was found to contribute to a lower consumption of carbohydrates and fats and maintaining energy balance, leading to weight loss.[100] Schruers et al. recorded an increase in the cortisol level in saliva of healthy volunteers after they used 5-HTP.[101] High cortisol levels correlate with negative effects on exercise in athletes. Cortisol also induces protein degradation and muscle damage.[102]

High consumption of preparations containing *G. simplicifolia*/5-HTP may cause euphoria. 5-HTP interacts with SSRIs and MAOIs, which may place the individual at risk of developing a serotonin syndrome. A potential analogous adverse interaction may occur by the combined use of *G. simplicifolia* and *H. perforatum*.[98,103] Hillebrand et al. reported that *G. simplicifolia* and 5-HTP were sold online as ingredients of 'designer drugs' or 'party pills' as an alternative to the synthetic psychoactive substance ecstasy.[104] A previous study demonstrated the occurrence of hypomania in a man using 5-HTP and adenosylmethionine.[105] Another study reported a case of hospitalization due to serotonin syndrome associated with *G. simplicifolia*/5-HTP supplementation.[106]

G. simplicifolia/5-HTP is present in the dietary supplements of the categories of fat burner, brain/cognitive booster, and hormone booster. According to the EFSA, seeds of *G. simplicifolia* are prohibited for use in dietary supplements and have been listed in the EFSA Compendium of Botanicals as a botanical substance of possible concern for human health.[60] The RASFF panel has received notification about 5-HTP as an unauthorized substance.[17]

Mucuna *sp.* (*L-DOPA*)

Seeds of *Mucuna pruriens* (Velvet bean) contain 3–6% levodopa (L-DOPA). L-DOPA is a nonprotein amino acid and a dopamine precursor in the human body. L-DOPA is not only on *Mucuna* sp. but also in many other plants, including some vegetables such as legumes. *M. pruriens* has been used in Indian ethnomedicine for centuries to improve sexual functions. *M. pruriens* has also been used as an antidote for scorpion bite. *Mucuna gigantea* is also known in Australia and Oceania not only as an aphrodisiac but also as an anthelmintic agent and soothing agent for insect bites. In Africa, *M. pruriens* seeds are used in traditional medicine as antidote for bite by poisonous snake species.[107,108]

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L-DOPA is present in therapeutic agents used for treating Parkinson's disease. The use of L-DOPA is associated with several central and peripheral side effects, including confusion, delirium, psychotic states, diarrhea, and vomiting.[109]

In Ayurvedic medicine, *M. pruriens* is recommended for treating Parkinson's disease. A previous study reported the limited effectiveness of using *M. pruriens* for relieving the symptoms of Parkinson's disease. The bioavailability of the active substance L-DOPA from *M. pruriens* seeds causes problems, which affects the effectiveness and safety of therapy of patients suffering from Parkinson's disease.[110–112]

In Ayurvedic medicine, *M. pruriens* is also used as an aphrodisiac to improve male potency and sexual performance. In scientific literature, Ahmad et al. and Shukla et al. reported that a 3-month supplementation of *M. pruriens* standardized for L-DOPA increased sperm motility and improved sperm parameters in infertile men.[113,114] In another study, an increase in testosterone concentration in infertile men was also observed.[115] Furthermore, it was proved that in a group of 15 physically active men, the acute supplementation of *M. pruriens* and *Chlorophytum borivilianum* caused an increase in the concentration of growth hormone in blood.[116]

The use of *Mucuna* sp. by humans has been associated with cases of psychotic disorders and acute food poisoning (vomiting and diarrhea) requiring hospitalization. The side effects of *Mucuna* sp. led to an increase in dopamine levels in the central and peripheral nervous system.[117–119] Studies have shown that *Mucuna* sp., in addition to L-DOPA, also contain other biologically active compounds, including tryptamine derivatives (bufotenin), beta-carboline derivatives, and alkaloids with tetrahydroisoquinoline structure. Bufotenin is an AChEI with hallucinogenic activity.[60,120]

M. pruriens standardized to L-DOPA occurs along with other stimulants in food supplements, such as caffeine and synephrine (*Citrus aurantium*), which may increase the risk of interactions dangerous to human health. In dietary supplements containing *M. pruriens*, a high content of L-DOPA was found in the range of 50–150 mg in a single or daily dose, depending on the product. These doses are close to (or equal to) the doses of L-DOPA found in prescription medicinal products.[121,122]

M. pruriens/L-DOPA has been recognized as a component of dietary supplements of the categories: pre-workout, brain/cognitive booster, and hormone booster. The whole plant of *Mucuna pruriens* has been listed as a toxic compound in the EFSA Compendium of Botanicals as a botanical substance of possible concern for human health.[60] The RASFF panel has received notification of *M. pruriens* as an unauthorized novel food ingredient.[17]

Evodia rutaecarpa (*Evodiamine and its Derivatives*)

Fruits of *Evodia rutaecarpa* contain indoloquinazoline-derived alkaloids, namely evodiamine and rutacarpine (also referred to as quinazoline-carboline derivatives in the literature).The fruits of *E. rutaecarpa* have been widely used in Traditional Chinese Medicine as an analgesic and cardiotonic agent and for treating gastrointestinal ailments.[123,124]

Few research studies have been conducted on the use of *E. rutaecarpa* in humans. Most of the studies involved *in vitro* tests or animal model experiments. Evodiamine and rutacarpine are agonists of vanilloid receptor (TRPV1) and may regulate thermogenesis. Previous studies have also shown sympathomimetic activity and inhibition of the enzyme acetylcholinesterase by evodiamine and rutacarpine.[123–125]

In a group of obese women, no weight loss effect was observed after *E. rutaecarpa* supplementation.[126] Furthermore, in a group of healthy and active men aged 18 to 30 years, *E. rutaecarpa* has been shown to be ineffective in inducing thermogenesis and increasing fat oxidation.[127]

E. rutaecarpa is an ingredient of dietary supplements of the categories of fat burners and pre-workout. *E. rutaecarpa* is listed in the EFSA Compendium of Botanicals as a botanical substance of possible concern for human health.[60] In the RASFF panel, *E. rutaecarpa* has been notified as an unauthorized novel food ingredient.[17]

Eurycoma longifolia (*Quassinoids*)

Eurycoma longifolia, also known as Longjack, Tongkat Ali, or Malaysian Ginseng, in Southeast Asia (Malaysia, Indonesia, and Thailand), has been used as an aphrodisiac and an antihyperglycemic agent. Roots of *E. longifolia* contain bioactive compounds known as quassinoids such as eurycomalactone, eurycomanone, and eurycomanol.[128]

In young, trained male cyclists, no improvement in cycling performance was observed after ingestion of *E. longifolia*. This ineffectiveness observed might be due to the low dose of *E. longifolia* in the range of 0.1 mg/100 ml of drink.[129,130] In a group of physically active men, the administration of *E. longifolia* at the dose

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100 mg per day for 5 weeks showed ergogenic activity: increased lean mass, reduced body fat, and increased muscle strength.[131] In healthy male recreational athletes, *E. longifolia* at the dose of 150 mg daily for 7 days showed no beneficial effects on endurance running performance.[132] In male athletes, *E. longifolia* ingested at the dose of 1.7 mg/kg body weight per day showed ergogenic benefits: induction of lipolysis, improvement in exercise performance, and enhancement of energy production.[133] *E. longifolia* has been proven to decrease cortisol levels and increase testosterone concentration in saliva of volunteers.[134]

E. longifolia is an ingredient in dietary supplements of the categories of pre-workout, fat burner, and hormone booster. In the EU, *E. longifolia* has been designated as an unauthorized novel food. The RASFF panel has been notified about *E. longifolia*, and this plant is listed in the EFSA Compendium of Botanicals as a botanical substance of possible concern for human health[17,60]

Salix *sp.* (*Salicin and its Derivatives*)

Willow belonging to the genus Salicaceae includes 500 different species of trees growing in Europe, North America, and Asia. The most frequently reported willow species in the scientific literature are *Salix alba* and *Salix purpurea*. Willow bark and willow bark extract have a long tradition of use in phytotherapy as an analgesic, antipyretic, and anti-inflammatory agent. Willow bark contains bioactive phenolic compounds from the group of phenolic acids. These are derivatives of salicylic acid, i.e., salicin and saligenin. In the 19th century, salicin was isolated from willow bark. Later, synthetic derivatives of salicylates were obtained, including acetylsalicylic acid (Aspirin).[135] Acetylsalicylic acid is a drug indicated for treating pain of various etiologies and is also used in cardiology as an anti-aggregation drug for thinning blood and inhibiting platelet clumping, because of which it plays an important role in the prevention of blood clots and embolism in blood vessels.[135] Willow bark/salicin shows anti-aggregation activity, but much weaker than that of acetylsalicylic acid.[135]

Willow bark has been listed as a raw material in the European and American pharmacopoeia and has also described in the monographs of the European Medicines Agency (EMA) and the European Scientific Cooperative on Phytotherapy (ESCOP). The monographs specify the minimum and maximum salicin content in willow bark in the range of 1.5–15% and the maximum daily oral dose of salicin not exceeding 240 mg.[135,136] Salicylates show analgesic, antipyretic, anti-inflammatory, and anti-aggregation activities, which results from the inhibition of cyclooxygenase-1 and cyclooxygenase-2 (COX-1 and COX-2, respectively). Inhibition of COX-1 and COX-2 activity blocks the synthesis of prostanoids: prostaglandins, prostacyclin, and thromboxane. The most commonly reported adverse effects associated with the use of salicylates are gastrointestinal disorders and allergic reactions in people allergic to salicylates.[135,136]

In early 1990, studies involving obese people were conducted, in which the effectiveness of the mixture of substances, namely 30–150 mg ephedrine, 150–200 mg caffeine, and 300–330 mg aspirin, in weight reduction was proven.[137–139] Dietary supplements (fat burner) were introduced to the distribution and containing a combination of active substances: salicylates, ephedrine (or synephrine), and caffeine. The combination of these substances aimed to increase thermogenesis and suppress the appetite, i.e., intensify cAMP production. The presence of natural or synthetic salicylates has been found in fat burners. Dietary supplements with willow bark extract contained a standardized amount of salicin or synthetic acetylsalicylic acid. However, not all fat burners with willow bark extract contained information about the exact amount of salicin (no standardization of the extract). In some dietary supplements, willow bark extract or salicin were part of a multicomponent mixture in which their exact amount was not specified.[137–139] Several cases of hepatotoxicity reported between 2003 and 2008 were associated with adverse effects of salicylates in dietary supplements (also containing ephedrine and caffeine) mainly used as slimming agents.[140,141] In one case report, after consuming a fat burner, a young woman required outpatient intervention due to an acute anaphylactic reaction, which manifested as hypotension; shortness of breath; and swelling of the throat, nose, and mouth. The cause of anaphylaxis was an allergy to salicylates contained in the dietary supplement.[142]

There are many reports of hospitalization in people taking weight loss supplements containing salicylates in combination with ephedrine/caffeine. Adverse cardiovascular symptoms associated with dietary supplements containing salicylates and ephedrine/caffeine are included in the section on prohibited ingredients (*Ephedra* sp.).

Studies have shown that salicylates do not affect exercise capacity. Salicylates have been approved by WADA for use by athletes during competition and in

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sports practice. The position of the IOC and WADA with regard to NSAIDs, including salicylates, informs that their use contributes to relieving muscle pain induced by exercise, during or after physical activity. By suppressing the sensation of pain, NSAIDs enable athletes to continue exercising.[13,143,144] In the EU, willow bark and willow bark extract are recognized as herbal medicinal products and are not included in dietary supplements.[145]

Berberis *sp.* (*Berberine*)

Berberine is an isoquinoline alkaloid present plant species such as *Berberis vulgaris*, *Berberis aristata*, *Berberis aquifolium*, *Hydrastis canadensis*, *Coptis chinensis*, and *Coptis japonica*. The root or bark of these plants was used in Ayurveda and in Traditional Chinese Medicine as an antibacterial, antiprotozoal, and antidiarrheal agent and was recommended for treating gastrological disorders.[146]

Berberine and selected species of *Berberis* are components of dietary supplements, whose effects help to burn fat tissue. Berberine can activate the AMPK enzyme (AMP-activated protein kinase) and inhibit PTP1B (protein tyrosine phosphatase-1B), which enables to regulate tissue sensitivity to insulin. Zhang et al. reported improvement in tissue sensitivity to insulin in patients suffering from type II diabetes who were supplemented with berberine. A meta-analysis conducted in 2012 showed that berberine use for 12 weeks in patients with diabetes improved all physiological markers of glucose and lipid levels.[147,148]

Berberine has been shown to lower total liver cholesterol and reduce triglyceride synthesis. The mechanism of pharmacological activity of this compound is based on its ability to inhibit the activity of HMG-CoA reductase (3-hydroxy-3-methylglutaryl-coenzyme A). In elderly patients with hypercholesterolemia, berberine was shown to affect the lipid profile, including lowering of total cholesterol, low-density lipoproteins, and triglycerides. Participants of the studies used a dietary supplement containing a combination of the following ingredients: berberine, red rice extract, and policosanol.[149,150] In more recent studies, a hypolipemic effect was observed in patients who used a preparation containing a combination of berberine, red rice extract, policosanol, and other ingredients.[151,152]

Kalman et al. proved that berberine relieved anxiety in healthy women. In another study, berberine reduced cortisol levels in a group of healthy men and women, which correlated with the effect on improving mood and reducing the feeling of stress and fatigue. In both these studies, the participants used a dietary supplement containing bark extract of *Magnolia officinalis* (honokiol) in addition to berberine.[153,154]

On the basis of latest findings, berberine is considered to be a safe substance. The reported adverse effects associated with the use of berberine are related to gastrointestinal disorders (abdominal pain, nausea, and vomiting). Studies have shown that berberine modulates the activity of cytochrome enzymes, i.e., it exhibits the features of an activator or inhibitor depending on the cytochrome isoform. Consequently, berberine may interfere with antidiabetic drugs, anticoagulants, statins, and antibiotics from the group of macrolides (risk of cardiotoxicity).[155,156]

The bark of *Berberis aristata* has been listed as a raw material in monograph in the British Pharmacopoeia. *Berberis* sp. is included in the EFSA Compendium of Botanicals as a botanical substance of possible concern for human health.[60] In the RASFF panel, berberine is recognized as an unauthorized novel food ingredient.[17]

Crocus sativus

Saffron, also known as crocus, is a popular ornamental plant. Because of its unique aroma, *Crocus sativus* is also used as a spice. It is one of the most expensive culinary additives in the world; this is because of the complex and time-consuming process for obtaining the raw material – crocus stigmas. Saffron originates from the Far East – Arab countries. *C. sativus* has already been described in traditional Persian medicine (Avicenna) as a plant with analgesic and emotion-regulating properties and as an aphrodisiac; it is also used as an abortifacient. The current use of saffron, e.g., in the food or cosmetic industry, results from its aromatic and coloring qualities.[72,157]

The results of *in vitro* and animal studies as well as human studies postulate the possibility of using saffron in regulating mood, reducing anxiety, controlling appetite, and alleviating the symptoms of premenstrual syndrome (PMS).[72,157] Gout et al. showed that in overweight women, supplementation with 176 mg of saffron extract a day affected satiety and reduced snacking, which might be helpful in reducing weight.[158] A previous study reported stress relief and mood improvement in healthy participants who consumed 28 mg of saffron extract per day.[159] Another study showed an ergogenic effect after a 10-day supplementation of 300 mg of saffron per day in a

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group of students. Saffron has been shown to shorten the reaction time and improve reflexes, alertness, and muscle coordination; this was thought to be associated with the effect of this material on the motor part in the cerebral cortex. There was also an increase in the strength of muscle contraction and an improvement in blood circulation, blood supply, and oxygenation to muscles and the brain.[160] Hosseinzadeh et al. showed that in women exercising, 4-week supplementation with saffron extract (30 mg per day) reduced the level of muscle damage markers: lactate dehydrogenase and creatine kinase; this might suggest a protective effect on muscle fibers. The results of these experiments indicate the antioxidant and anti-inflammatory effects of saffron.[161] The protective effect of saffron on muscles is described by Meamarbashi et al. Saffron used for 10 consecutive days was found to be more effective than indomethacin in reducing symptoms of delayed onset muscle soreness (DOMS).[162] Another study proved that an acute intake of 300 mg of powdered saffron to young and healthy volunteers can be effective in increasing anaerobic exercise capacity.[163]

Crocus contains phytochemicals from the group of monoterpenes – safranal; flavonoids – quercetin and kaempferol; and carotenoids – crocin and crocetin. The antidepressant and appetite-regulating effects are related to the influence of safranal and crocin on the neurotransmission of catecholamines in the CNS. Safranal has a greater affinity for inhibiting serotonin reuptake, and crocin affects blocking noradrenaline and dopamine reuptake.[72,157] Studies on volunteers have shown adverse effects of saffron, i.e., gastric disorders, nervousness, dry mouth, and increased sweating. Information on hematological changes in blood is a cause of concern, and safranal is considered to be a compound with the greatest toxic potential. A daily dose of more than 1000 mg of saffron in humans may cause vomiting, diarrhea, and bleeding. These adverse reactions were also observed during supplementation of lower doses of approximately 60–400 mg of crocus.[157] Hemodynamic disorders, i.e., reduction in blood pressure and changes in blood count, were observed in healthy volunteers during 7-day supplementation with 200 mg or 400 mg of saffron. Two women showed abnormal vaginal (uterine) bleeding.[164] Safarinejad et al. showed that long-term supplementation of 60 mg per day of saffron in men correlated with a decrease in the count of erythrocytes, leukocytes, and platelets as well as with lowering of blood pressure. Studies also showed the effect of saffron on appetite change.[165] The most commonly used saffron extract in dietary supplements is the one standardized to 5% crocin. In addition to crocin standardization, dietary supplements with quantified safranal content are also available. On the basis of published literature, it can be concluded that a dose of 30 mg of saffron per day is well tolerated by the human body and does not cause any adverse events.[166,167]

Crocus sativus is not listed in the EFSA novel food catalogue or EFSA Compendium of Botanicals, and it is recognized as an authorized ingredient in dietary supplements.[17,60]

Tribulus terrestris (*Saponins*)

Tribulus terrestris was used in ethnomedicine in China, India, and Eastern Europe (Bulgaria) as an aphrodisiac, appetite stimulant, and supplement for proper functioning of the urinary and cardiovascular systems.[168,169] *T. terrestris* functions as a component of the dietary supplements called *testosterone boosters*, which are advertised as preparations that improve physical exercise capacity, strengthen men's potency, and enhance sexual performance. *T. terrestris* gained popularity as food supplements after Bulgarian athletes won Olympic medals at the Atlanta Olympics in 1996. At that time, *T. terrestris* was intensively promoted as an ingredient in Tribestan preparation, which was produced by a company from Bulgaria. A significant amount of research on the effect of *T. terrestris* on testosterone levels and improvement in sexual function was carried out by Bulgarian scientists in the 1980s.[168,169] In the subsequent years, it was shown that the enhanced performance of Bulgarian athletes was due to the use of AAS, not *T. terrestris.* Bulgarian athletes were banned from participating in the Beijing Olympic Games in 2008. There is no scientifically proven evidence regarding the effect of *T. terrestris* on increasing testosterone production in men or in improving strength and physical endurance. Only few studies such as Milasius et al. have reported an increase in physical strength and testosterone levels in young men supplemented with *T. terrestris.*[170] In another study, older men with age-related androgen deficiency showed an improvement in sexual function (erection) and an increase in testosterone level.[171] Salgado et al. also showed an increase in testosterone concentration in men diagnosed to have infertility and were supplemented with *T. terrestris.*[172] Antonio et al., Van Eenoo et al., Poprzecki et al., and Rogerson et al. confirmed that the supplementation of *T. terrestris* did not show a significant effect on physical performance and testosterone level in healthy and physically active humans.[173–176] Another hypothesis

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is related to the effect of *T. terrestris* on enhancement of nitric oxide production, which is associated with the effect of supporting the spermatogenesis process and increasing exercise capacity.[177] There are known cases of identification of prohibited components such as AAS or hormone precursors in dietary supplements containing *T. terrestris*. The RASFF panel has received notifications about incidents of detection of tetrahydrocannabinol (THC) in dietary supplements containing *T. terrestris*.[17,178] *T. terrestris* food supplements available in the market are characterized by a high content of steroid and triterpene saponins (dioscin, diosgenin, and protodioscin). The allowed amount of *T. terrestris* extract in dietary supplements in the member states of the EU has not been completely specified.[179] The fruit of *T. terrestris* is listed in the British Pharmacopoeia. Food supplements containing *T. terrestris* extract are available in the market, in which the content of saponins was determined to be too high. The geographic region from which the native raw material of *T. terrestris* originates has an influence on the content of the saponin fraction.[179,180] A case of gynecomastia has been reported in a man who used dietary supplements with *T. terrestris*, which confirms the effect of *T. terrestris* on the hormonal balance of the body and the possibility of its regulation.[181] Another study reported the nephrotoxic and hepatotoxic effects observed during supplementation with *T. terrestris*.[182] Acute kidney failure and hyperbilirubinemia associated with *T. terrestris* supplementation are also reported.[183] A case report cautioned about the occurrence of cerebral venous thrombosis in a 37-year-old man after supplementation with *T. terrestris*.[184]

T. terrestris has been detected in the dietary supplements of the category of hormone/testosterone booster. In the EU, *T. terrestris* is recognized as an authorized novel food and can be used in dietary supplements, with some restrictions regarding the dose and amount of saponins.[17,179] *T. terrestris* is listed in the EFSA Compendium of Botanicals because toxicity to the CNS and liver was observed in experimental animals.[60]

2.2. Synthetic ingredients

Dimethylaminoethanol and its Derivatives

Dimethylaminoethanol (DMAE), dimethylethanolamine or deanol, is a hydroxy derivative of choline. After ingestion and absorption into the bloodstream, DMAE crosses the blood–brain barrier and becomes a precursor of acetylcholine in the CNS.[47,185] DMAE in the form of salts or esters has long been used as a component of many therapeutic agents. The preparation containing this structure, i.e., *Centrophenoxin*, was used in geriatrics, particularly in patients with dementia or stroke.[186] Subsequent preparations, namely *Risatarun* and *Pirisudanol*, are examples of drugs that have been used in pediatrics for the treatment of bedwetting and attention-deficit hyperactivity disorder (ADHD) in children. DMAE metabolism in the human body can lead to the formation of carcinogenic nitrosamines. DMAE has also been found to be teratogenic.[47,187]

On the basis of the current scientific literature, the use of racetam nootropics in dietary supplements is known. Among racetams, most substances are registered as medicines, e.g., piracetam in Poland. The distribution also includes dietary supplements containing other derivatives that show greater pharmacological activity than the parent piracetam. Piracetam derivatives used in dietary supplements include oxiracetam, aniracetam, phenylpiracetam, and noopept.[188]

DMAE and racetams have been detected in the dietary supplements of the category of brain/cognitive booster and have often been combined with other substances showing activity on the CNS, such as *Vinca* minor (vincamine), *Huperzia serrata* (huperzine), *Piper methysticum*, *Griffonia simplicifolia*, and *Mucuna* sp. (L-DOPA). DMAE has also been found in pre-workout supplements in combination with *Corynanthe yohimbe* (yohimbine), *Rauwolfia* sp., *Sceletium tortuosum*, caffeine, agmatine, or synephrine.

DMAE and racetams are designated as unauthorized novel food ingredients in dietary supplements in the EU.[16,17]

Phenibut

Phenibut (4-amino-3-phenylbutyric acid or γ-amino-β-phenylbutyric acid) is a synthetic psychotropic drug that was developed in the early 1960s in the Soviet Union, where phenibut was used as an anxiolytic, sedative, and hypnotic drug as well as a nootropic agent. Phenibut is still used in pharmacotherapy in Russia, Ukraine, and Latvia in premedication as a sedative and reliever of anxiety related to surgery, in the treatment of sleep disorders and anxiety disorders, for alleviation of withdrawal symptoms in alcohol-dependent individuals, and for treating dyslexia in children. The trade names of drugs containing phenibut are Anvifen, Bifren, Citrocard, and Noofen.[189] It should be noted that phenibut was included in medicines provided Russian cosmonauts during a flight into space, as a sedative and for simultaneously improving concentration and cognitive functions.[190]

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In terms of chemical structure, phenibut is a phenyl analog of γ-aminobutyric acid (GABA) – a neurotransmitter that suppresses the activity of the CNS. The phenyl substituent attached to the GABA aliphatic chain increases the lipophilicity of the phenibut molecule, which enables it to penetrate the blood–brain barrier and induce the pharmacological effect after oral administration.[189,190]

Phenibut has a chemical structure similar to that of two other GABA structural analogs, namely baclofen and gabapentin. Baclofen and gabapentin have been approved for pharmacotherapy in the US and EU. Baclofen differs from phenibut by a chlorine atom in the *para* position of the phenyl substituent. Baclofen is a myorelaxant drug that reduces the tension of skeletal muscles in various disease states, e.g., multiple sclerosis, cerebral palsy, and post-traumatic rehabilitation. Gabapentin has a cyclohexyl substituent (cyclohexane ring) in the place of the phenyl substituent in relation to the chemical structure of phenibut. Gabapentin is a drug indicated in the treatment of partial epileptic seizures with secondary generalized seizures, in the treatment of neuropathic pain, and as an adjunct in bipolar disorder. Phenibut has a chemical structure similar to one more structure, namely gamma-hydroxybutyric acid (GHB), which is used for finding criminal evidence.[189–191]

Phenibut has a greater affinity for the GABA-B receptor. Phenibut stimulates dopamine receptors and increases serotonin transmission in the CNS. Phenibut binds to the α2-δ subunit of the voltage-dependent calcium channels (VDCCs). Phenibut has been shown to reduce the activity of VDCCs.[189,192,193] It contains one asymmetric carbon atom and is a racemic mixture of the R and S enantiomers. Phenibut with R configuration shows pharmacological activity.[194] Most of the research on phenibut has been conducted in the Soviet Union/Russia. It has been proven that phenibut improves cerebral circulation in swimmers.[195]

Phenibut is an ingredient of dietary supplements in the brain/cognitive booster category sold online. It is commonly referred to as 'smart drug.' Information on dietary supplements containing phenibut was first available in 2005 and was posted on an Internet portal for athletes.[196]

In 2008, a case of a young man was reported who developed psychotic symptoms after discontinuing phenibut.[197] The first warnings from government organizations about phenibut, a new psychoactive substance in dietary supplements, appeared in 2012.[198]

Phenibut dietary supplements are often used in an uncontrolled manner, and doses of phenibut delivered to the body exceed the therapeutic doses prescribed for this drug by Russian authorities, i.e., 250–2000 mg per day. High doses of phenibut can lead to intoxication and adverse reactions. The acute toxicity of phenibut is manifested by alternating episodes of disturbance of consciousness, sedation, lethargy, depersonalization, dementia, hallucination and episodes of psychomotor agitation, aggression, dystonia, hypothermia, tachycardia, and serotonin syndrome.[199]

Another case of a male patient has been reported who developed dissociative symptoms after an overdose of phenibut, preceded by insomnia for last four days. In the course of the patient's treatment, psychotic symptoms developed suggesting phenibut withdrawal symptoms.[200,201]

Easy access to dietary supplements containing phenibut results in its abuse for recreational purposes as a euphoric or intoxicating substance. Phenibut has an addictive potential with the development of tolerance and a specific set of withdrawal symptoms. Phenibut tolerance develops relatively quickly after 1–2 weeks of use.[202] Symptoms of withdrawal occur 2–4 h after consuming the last dose of phenibut. Symptoms of discontinuation of phenibut include visual and auditory hallucinations, insomnia, anxiety, irritability, hypersensitivity to external stimuli (light, sound, and temperature), muscle pain and tremors, weakness, and decreased appetite.[201]

A systematic review conducted in 2020 examined the safety of phenibut based on clinical trials and case reports. It has been established that at therapeutic doses of 250–2000 mg per day, phenibut is a well-tolerated drug with few side effects. The review highlights the problem of phenibut, a drug that has been approved in some Eastern European countries and is an unauthorized ingredient in dietary supplements sold in the US, Australia, and the EU. The published article addresses the quality of phenibut used in the production of dietary supplements as well as the problem of phenibut abuse in high doses for recreational purposes as a euphoric/intoxicating substance by people who have consumed other psychoactive substances in the past or people looking for alternative and readily available (legally purchased) psychoactive drugs.[202]

In the EU, phenibut is not registered as a medicinal product, but it is considered as a novel psychoactive substance (NPS). Phenibut is not included in the RASFF reports and in the EU Novel Food Catalogue.[16,17]

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Agmatine

Agmatine is a biogenic amine that is chemically derived from guanidine. In the human body, agmatine is produced by the decarboxylation reaction of the amino acid L-arginine. Agmatine is an intermediate in polyamine biosynthesis. Agmatine is found in some foods, especially fermented foods such as wine and beer. Agmatine is present in small amounts under physiological conditions in the human body. The presence of synthetic agmatine sulfate was found in dietary supplements.[203,204]

Most of the studies on the use of agmatine were performed in *in vitro* tests or in an animal model. Agmatine has a broad biological activity profile. It is an antagonist of NMDA and nicotinic cholinergic receptors. It has been shown to be an agonist of serotoninergic, imidazoline, and α_2-adrenergic receptors. The stimulation of imidazoline receptors correlates with the neurotransmission of endogenous opioids. Agmatine regulates the activity of calcium channels and is an inhibitor of synthase nitric oxide (NO); it increases the concentration of NO.[203,204]

Scientific studies indicate the synergism of agmatine with analgesic drugs from the opioid group in terms of analgesic activity. Agmatine reduces the addictive potential of opioids and limits the development of tolerance.[204,205,206]

Few clinical trials have been performed on the use of agmatine. In a clinical trial in patients with neuropathic pain, agmatine was shown to have analgesic activity.[205] In a study on patients with depression, the application of agmatine at the dose of 2–3 g per day led to remission of depression symptoms. A report confirmed the safety of high daily doses of agmatine at 2.67 g consumed over a period of 5 years.[206]

There are no scientific studies in athletes confirming the ergogenic activity of agmatine. According to the information of the International Society of Sports Nutrition (ISSN), agmatine is placed in category III of ingredients influencing the growth of muscle mass, for which there is little scientific evidence indicating the effectiveness of its use.[11,12]

Agmatine is an ingredient of dietary supplements in the pre-workout category. Furthermore, it has been notified as an unauthorized novel food ingredient in the RASFF panel and the EFSA Novel Food Catalogue.[16,17]

3. Ingredients Prohibited by WADA

3.1. *Natural Ingredients and Isolated Active Substances*

Ephedra *sp.* (*Ephedrine*)

Ephedrine is an alkaloid from the group of phenylethylamine (PEA) derivative found in the herb of Chinese Ephedra (*Ephedra sinica*) and *Ephedra equisetina*. Ephedrine works by stimulating the central and peripheral nervous system.[207–210] It is a sympathomimetic amine, which increases the release and inhibition of the reuptake of catecholamines, including most of all, norepinephrine in synaptic clefts. Ephedrine is a nonselective agonist of α-adrenergic and β-adrenergic receptors. The activation of α_1-adrenergic receptor leads to vasoconstriction and increase in blood pressure. Stimulation of β_1-adrenergic receptors located in the heart muscle results in the acceleration of heart rate. Ephedrine is also an agonist of β_2-adrenergic receptor located in the bronchi, which results in the bronchodilator effect. Ephedrine is a component of therapeutic agents used for bronchial spastic conditions. *Ephedra* sp. or ephedrine has also been used in dietary supplements of the fat burners or thermogenic category.[47,207]

In a group of obese adults, a significant decrease in body weight and a reduction in body fat was found during the use of the Xenadrine dietary supplement containing 20 mg of ephedrine, 5 mg of synephrine, 200 mg of caffeine, and 15 mg of salicin. The safety of this combination of active substances was found in studies conducted on human subjects. In an experiment conducted in 2002, the safety of Xenadrine supplementation in obese volunteers was shown, but 2 years later, a correction to the conducted studies was published, in which adverse effects such as dyspnea, chest pain, overagitation, and sleep disturbances were reported.[208]

The first publications describing adverse events in humans consuming ephedrine-containing dietary supplements were reported in 1996. Finally, in an experiment conducted in 2005 on a group of healthy volunteers, it was proved that the combination of substances containing 20 mg of ephedrine, 5 mg of synephrine, 200 mg of caffeine, and 15 mg of salicin contributes to hemodynamic disorders: acceleration of the heart rate, increase in blood pressure, and reduction of potassium in blood.[209] Several cases of hospitalization of people using ephedrine supplements or the combination of substances such as ephedrine, salicylates (or aspirin), and caffeine have been reported. Because of these measures, outpatient

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intervention was necessary due to serious cardiovascular complications such as tachycardia, sharp increase in blood pressure, and/or thermoregulation disorders – hyperthermia. Serious cardiological incidents that have been demonstrated include myocardial infarction and neurological complications such as seizures. A confirmed case was a heatstroke in a US military person after consumption of the dietary supplement Xenadrine. The use of the dietary supplement Xenadrine resulted in the death of a baseball player from heat stroke. Cases of cardiomyopathy (including 1 death) associated with the consumption of a fat burner containing salicylates, ephedrine, and caffeine have been reported in young people.[209,210]

Information on the detection of unauthorized stimulants in dietary supplements containing ephedrine is presented in publications. The research was aimed at informing and warning the sports environment (before the upcoming Olympic Games) about the potential risk of detecting doping in competitors using dietary supplements.[14] Ephedrine is recognized on the WADA Prohibited List in group S6 – Stimulants:[13] Ephedrine was withdrawn as a component of dietary supplements in the United States by the FDA in 2004. In the EU, *Ephedra* sp. or ephedrine cannot be included in dietary supplements, and the decision regarding this was made in 2013.

Citrus *sp.* (*Synephrine and its Derivatives*)

Synephrine is a natural alkaloid from the PEA derivative group and is found in citrus fruits, e.g., bitter orange (*Citrus aurantium*). The most commonly used species in dietary supplements is *C. aurantium.* In addition to synephrine, *C. aurantium* contains the presence of other active compounds like synephrine derivatives such as methylsynephrine (oxilofrine), octopamine, and hordenine. Synephrine its sympathomimetic amine, agonist of the α_1-adrenergic and β_3-adrenergic receptor. Synephrine, also interacts with trace amine-associated receptor 1 (TAAR1). methylsynephrine shows greater α-adrenergic receptor activity than synephrine. Octopamine and synephrine have similar affinity to α-adrenergic receptor, but octopamine shows greater activity toward the β-adrenergic receptor than synephrine.[211,212]

Synephrine is a close structural analog of ephedrine. Opinions are divided on the efficacy and safety of the use of synephrine and *C. aurantium* by humans. *C. aurantium*/synephrine was demonstrated to be the most popular substitute replacing ephedrine, which was withdrawn from use in food supplements. *C. aurantium*/synephrine use in combination with other stimulants such as caffeine has been associated with many adverse reactions in the cardiovascular system.[211,212]

It is not entirely clear whether only the *para*-synephrine isomer is present in *C. aurantium.* It is known that *meta*-synephrine is a synthetic isomer present in medications indicated to decongest nose in colds or allergies. Most scientific studies indicate that only the *para*-synephrine isomer occurs naturally in *C. aurantium.* Dietary supplements with *C. aurantium* have also been found to contain the isomer *meta*-synephrine. This may indicate the addition of synthetic *meta*-synephrine to food supplements. Many dietary supplements have been shown to contain extracts with a high standardization level of synephrine in the range of 10–90%. However, in the natural sources of synephrine from fruits of *Citrus* sp., such a high concentration of this alkaloid has not been detected.[211,212]

The issue of the occurrence of bitter orange extract in dietary supplements is additionally complicated by the fact that some of the food supplements do not have a specific standardization and exact content of synephrine. Synephrine is included in the WADA Prohibited List of doping agents as a monitored substance in the group S6 – Stimulants.[13] A review of scientific literature showed that synephrine derivatives such as octopamine and methylsynephrine have been identified in dietary supplements. Octopamine and methylsynephrine are recognized not only as monitored substance but also as doping agents from the group S6 – Stimulants. An example of disqualification for unintended doping in 2013 was the female sprinter Sherone Simpson, as the anti-doping tests performed on her showed the presence of methylsynephrine, which came from the food supplement she was using.[211–213]

Isopropyloctopamine (also known as isopropylnorsynephrine, betaphrine, or deterenol) is a synthetic compound found in dietary supplements in the categories of fat burner and pre-workout. Isopropyloctopamine is a selective agonist of β-adrenergic receptor; it has a high affinity for β_3-adrenergic receptor and influence lipolysis. In the dietary supplement, isopropyloctopamine was present in the amount of 40–60 mg. The use of dietary supplements containing isopropyloctopamine was correlated with 11 reports of adverse cardiovascular events in the Netherlands. Isopropyloctopamine is not included in the WADA Prohibited List. However, WADA specifies that '*and other substances with a similar chemical structure*

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or similar biological effect(s) are also prohibited'. As described in the following sections, the excessive use of certain substances present in dietary supplements by professional athletes has forced WADA to adapt and update its regulations to the current anti-doping needs.[13,211–213]

The presence of *C. aurantium* or synephrine was found in dietary supplements of the categories of fat burner and pre-workout. In the EU, *C. aurantium* is designated as an authorized ingredient in food supplements, but with restrictions on the maximum concentration of synephrine up to 5%. *Citrus* sp. are listed in the EFSA Compendium of Botanicals because of the content of alkaloids (synephrine) in unripe whole fruit (2.28 mg/g) and pericarp (3.27 mg/g).[60] In the RASFF panel, synephrine, oxilofrine, octopamine, and isopropyloctopamine have been notified as unauthorized substances.[17]

3.2. *Synthetic Ingredients*

After the withdrawal of *Ephedra* sp./ephedrine, synthetic substitutes for *Ephedra* sp. were identified in dietary supplements. These ingredients, namely phentermine, fenfluramine, and sibutramine are classified as medications and so-called anorectic agents and used in pharmacotherapy for obesity. Extreme cases involve the detection of the derivatives of the drug amphetamine – methamphetamine and cathine – in dietary supplements. Other stimulants analytically labeled in controlled food supplements include pseudoephedrine – a popular ingredient in over-the-counter (OTC) medications for rhinitis and cold.[13–15]

DMAA and its Derivatives

According to a study in 2006, a new psychostimulatory constituent – dimethylamylamine (DMAA) (*Figure 1*) – was found in dietary supplements.[214,215] In the 1940s, Eli Lilly Company introduced DMAA in medical treatment due to its sympathomimetic activity – rendering the effect of nasal mucosal vasoconstriction. DMAA has been used as an aerosol to treat nasal obstruction and inflammation. In the 1970s, DMAA was removed from the pharmaceutical market. Forty years later, DMAA has returned to the market, not in pharmaceuticals, but in dietary supplements.[214,215]

Adverse cardiovascular events associated with the consumption of dietary supplements containing DMAA have been reported, including cerebral hemorrhage and death.[214,215] Cases of hepatotoxicity have been associated with the fat burner OxyElite Pro™, which contained DMAA. Hepatotoxicity was the reason for the product being removed from the market. It has now been established that liver damage is associated, not with DMAA, but with the presence of a natural ingredient, Indian plant *Aele marmelos*, which contains the alkaloid aegeline. Aegeline is a structural derivative of PEA.[216]

DMAA in dietary supplements has often been labeled with different names, including 1,3-dimethylpentylamine, geranium, Geranamine™, and geranaburn. In other products, a herbal ingredient was declared, e.g., *Pelargonium graveolens* and *Dendrobium nobile*, without providing any information about the presence of DMAA in order to 'camouflage' and make it difficult to identify the parent active ingredient.[214,215,217] *P. graveolens* (geranium) and *D. nobile* are recognized as an unauthorized novel food in the EU.[16,17]

In 2010, DMAA was placed in the Prohibited List of doping agents from the group of S6 – Stimulants. In 2012, in USA, the FDA questioned the use of DMAA. In the same year in the EU, the same decision was made by the EFSA.[214,215,218] The RASFF panel has received notification of DMAA as an unauthorized substance.[17]

Following the questioning of the use of DMAA, a new derivative was identified in dietary supplements – dimethylbutylamine (DMBA) (*Figure 1*). DMBA was introduced for use in food supplements, despite the lack of research on its effects in the human body. The presence of DMBA was found by analytical studies of dietary supplements containing Chinese tea extract. DMBA was placed in the WADA Prohibited List of doping agents in 2018.[13,214,219] In the EU and RASFF panel, DMBA has been notified as an unauthorized substance.[17]

Among other psychostimulants, the presence of which has been shown in preparations based on scientific research, we should mention PEA derivatives. β-Methylphenylethylamine (BMPEA) was first reported as a confirmed component of dietary supplements in 2009.[220] BMPEA is regarded as an unauthorized substance in dietary supplements in the EU.[17]

N-α-Diethylphenylethylamine (DEPEA) has been found in dietary supplements with the declared content of the plant raw ingredient – *Dendrobium*. On the basis of scientific literature, it has been shown that in 2012, a new synthetic psychostimulant substance was identified in the studied dietary supplements: *N*,*N*-dimethyl-2-phenylpropan-1-amine (DMPPA), which was detected in products containing the plant extract of *Acacia rigidula*.[221] PEA and its derivatives are recognized by WADA as doping agents from group S6

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– Stimulants. *A. rigidula* is recognized as an unauthorized novel food in the EU.[13,16,17]

In 2017, another new psychostimulant component from the group of aliphatic amines was found in the dietary supplements selected for research – dimethylhexylamine (DMHA) (*Figure 1*).

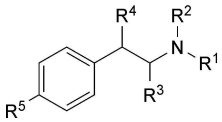
The interchangeably used names of DMHA include octodrine, 2-aminoisoheptane, and 2-amino-6-methylheptane. DMHA is a structural analog of the previously mentioned DMAA and DMBA. Analytical studies have revealed the presence of DMHA in food supplements containing the plant raw material *Aconitum kusnezoffii* or *Kigelia africana* – being the matrix for synthetic octodrine. In Europe, octodrine has been used in the form of oral tablets to treat orthostatic hypotension, while in the USA, octodrine has been used as an inhalator for treating asthma. DMHA has been placed in the Prohibited List of doping agents from group S6 – Stimulants since 2020.[214,222] In the RASFF panel, DMHA has been notified as an unauthorized substance.[17]

Based on the analyzed scientific literature, it can be concluded that since 2006, a greater number of new plant-based components have been used in dietary supplements, especially in those targeted to people who show increased physical activity, and their components have not been medically confirmed. These raw materials include *Senegalia berlandieri*, *Eria jarensis*, *Kigelia africana*, *Dendrobium nobile*, *Pelargonium graveolens*, and *Acacia rigidula*. The effect of the plant species used in dietary supplements was explained by contamination of psychostimulants from the PEA group or aliphatic amines. Some plants may contain naturally occurring PEA or aliphatic amines, but not in the concentration found in research on selected products. Dietary supplement producers often mention only the botanical name of the plant in their product descriptions, without giving information to consumers about the stimulants present within.[214,217,222]

The alkyl group in R^1 and/or R^2 position increases the lipophilicity of the molecule, enhances the psychostimulatory effect, extends the duration of effect, and is responsible for the nonselective stimulation of α- and β-adrenergic receptors. A larger group in R^1 substituent increases the toxicity of the compound, e.g., aegeline (*Table 1*).[47,51,223]

The alkyl group in R^3 substituent reduces enzymatic degradation by monoamine oxidase (MAO), extends the duration of action, and increases the psychostimulatory effect. The alkyl group in R^4 position determines the effect on the transport of serotonin and dopamine, which correlates with the dysleptic effect (BMPEA and DMPPA). The hydroxy group in R^4 position is responsible for the affinity for α- and β-adrenergic receptors. Alkyl or alkoxy substituent in R^5 position in the aromatic ring protects the structure from inactivation by catechol-*O*-methyltransferase (COMT), provides greater affinity for the serotonergic and dopaminergic system than for the adrenergic system, shifts the action profile from the CNS stimulation toward hallucinogenic effect, increases lipophilicity of the molecule and penetration through biological membranes, and increases the toxicity toward the human body. The hydroxy group in position R^5 reduces the lipophilicity of the molecule and penetration through the blood–brain barrier, weakens the psychostimulatory effects, and determines the affinity for α- and β-adrenergic receptors.[47,223]

Table 1. General structure of phenylethylamine and substituents affecting activity/toxicity of derivatives.



Chemical compound	R^1	R^2	R^3	R^4	R^5
Ephedrine	CH_3	H	CH_3	OH	H
Synephrine	CH_3	H	H	OH	OH
Methylsynephrine	CH_3	H	CH_3	OH	OH
Octopamine	H	H	H	OH	OH
Isopropyloctopamine	$CH(CH_3)_2$	H	H	OH	OH
BMPEA	H	H	H	CH_3	H
DEPEA	CH_2CH_3	H	CH_2CH_3	H	H
DMPPA	CH_3	CH_3	H	CH_3	H
Aegeline	$COCH{=}CHC_6H_5$	H	H	OH	OCH_3

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Higenamine

Higenamine, also called norcoclaurine, is an isoquinoline alkaloid (*Figure 1*) that occurs in some plant species, including *Nelumbo nucifera*, *Nandina domestica*, *Magnolia salicifolia*, and *Aconitum japonicum*. In Traditional Chinese Medicine, *Nandina domestica* has been recommended for respiratory diseases to relieve breathlessness. The concentration of higenamine in natural sources from plants is low, and therefore, a method of preparing synthetic higenamine was developed.[224] Therefore, an overview of higenamine is presented in this section of synthetic ingredients.

Higenamine is an agonist of β_1-adrenergic receptors located in the heart muscle and β_2-adrenergic receptors located in the bronchi. It was found that higenamine influences cardiac chronotropism and inotropism, and broadens the bronchi, which may affect physical exercise capacity. The stimulation of β-adrenergic receptors also correlates with the severity of the reduction of adipose tissue. Early publications from the 1980s in China describe the effect of higenamine on the efficiency of the left ventricle.[224] The presence of higenamine in dietary supplements was first reported in 2012. Feng et al. confirm the pharmacokinetic and pharmacodynamic profile of higenamine with (controversial) combination of substances, caffeine and yohimbine, and its safety in healthy volunteers.[225] Another research article showed that the combination of higenamine, caffeine, and yohimbine stimulates lipolysis in healthy and physically active humans.[226] The safety of higenamine in doses of 50 mg/day and 150 mg/day, when used for 8 weeks in a group of young, healthy men, has been confirmed.[227]

Jeter et al., reported the case of muscle rhabdomyolysis in a young man using a dietary supplement containing higenamine.[228]

Cases of the use of higenamine in professional sports were reported in 2016, including those of a Polish athlete and a UEFA Europe League football player. Since 2017, higenamine is placed in the Prohibited List of WADA doping agents in group S3, β_2-agonists.[13] Further examples of sanctions for doping caused by the detection of higenamine involve American athletes in 2018. Most of these cases of higenamine determination during doping control were correlated with the use of food supplements by athletes.[229]

Higenamine is found in dietary supplements from the pre-workout and fat burner categories, and it is often used in combination with caffeine, yohimbine, agmatine, synephrine, DMAA, DMBA, or DMHA. In the EU, higenamine has been notified as an unauthorized substance.[17] Sometimes, dietary supplements are specified to contain a plant extract with a standardized amount of higenamine. *Nelumbo nucifera* is recognized as an unauthorized ingredient in the RASFF panel and has the status of novel food in the EFSA Novel Food Catalogue.[16,17] *Magnolia salicifolia* and *Aconitum japonicum* are listed in the EFSA Compendium of Botanicals. *Nandina domestica* is not notified in the RASFF panel, the EFSA Novel Food Catalogue, and the EFSA Compendium of Botanicals.[16,17,60]

Clenbuterol

Clenbuterol jest is a long-acting and selective agonist β_2-adrenergic receptor. Clenbuterol was synthesized in 1967 by the pharmaceutical company Boehringer Ingelheim. It was introduced in pharmacotherapy in 1976 in Germany as a bronchodilator indicated for treating asthma. Since 1980, clenbuterol has been used as a veterinary medicine in horses for treating respiratory diseases with narrowing of the airways due to bronchospasm, e.g., chronic obstructive pulmonary disease in horses. Clenbuterol was also used as a doping agent for race horses.[230,231]

Some studies of clenbuterol have been conducted in humans. Bronchodilator activity has been confirmed in patients with asthma. The bioavailability and pharmacokinetic parameters of clenbuterol are similar to those of ambroxol.[230,231] It has been proven that clenbuterol at the dose of 40 μg/day can help maintain strength and muscle mass in men and also accelerate the rehabilitation process after an injury.[232] In a group of patients with heart failure, 80 μg per day of clenbuterol was shown to increase lean mass and muscle strength. No improvement was observed in endurance or exercise performance.[233] In young healthy men, confirmed enhancement of thermogenesis, increase in fat oxidation, and increase in metabolic rate were observed. In addition, clenbuterol induced mTOR phosphorylation in muscles; this may account for anabolic activity and muscle hypertrophy.[234]

Clenbuterol is used by professional and amateur athletes as an ergogenic aid to physical performance. It contributes to the growth of muscle mass (anabolism) with simultaneous lipolytic activity, as it reduces adipose tissue. Clenbuterol also affects aerobic capacity due to its broncholytic effect. It gained popularity among professional and amateur bodybuilders in the early 1990s. The strong anabolic effect of clenbuterol made it known as a substitute or alternative to

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anabolic androgenic steroids. Clenbuterol is placed in the WADA list of doping agents in group S1 – Anabolic Agents.[13] There are known cases of detection of clenbuterol in professional athletes during anti-doping control.[231–235]

The first detection of clenbuterol in dietary supplements was reported in Germany in 2008. The case involved a man who presented with disturbing muscle tremors to the outpatient clinic after consuming a dietary supplement – a fat burner. Clenbuterol concentration of 2 ng/ml was found in the urine of the patient. In the analyzed fat burners, clenbuterol was determined in the amount of 30 μg per tablet. According to the manufacturer's information on the label, the fat burner was recommended to be used as 3 tablets per day; the daily dose of clenbuterol delivered to the body, i.e., 90 μg, exceeded the therapeutic indications in the range of 40–60 μg per day.[236]

There are known serious cases of adverse effects resulting from the use of clenbuterol in humans, such as tachycardia or myocardial infarction.[237]

Clenbuterol has not yet been approved for pharmacotherapy by the FDA in the USA. In some countries in Europe, clenbuterol has been approved as a veterinary drug. Clenbuterol, also is used by breeders of slaughter animals to increase weight of the animals. In most countries, its use in animal husbandry is prohibited because it can accumulate in meat, which when consumed by humans leads to the detection of clenbuterol. Such cases have been described among professional athletes. There are reports in the RASFF panel about meat containing the unauthorized substance – clenbuterol.[17,238]

Anabolic Androgenic Steroids

The first studies of dietary supplements were performed in the late 1990s. A previous study indicated the presence of steroid compounds such as dehydroepiandrosterone (DHEA) in dietary supplements. Other studies indicated that hormone precursors and anabolic androgenic steroids (AAS) were present in food supplements, including testosterone, norandrosterone, and androstenedione derivatives.[239] In IOC international research in 2001–2004, dietary supplements from 13 different European countries were analyzed, and the presence of AAS was found in 15% of approximately 600 verified products.[240]

For the first time in 2002, dietary supplements were labeled with a new steroid group (designer steroids) such as prostanazole, metasterone, and formestane. In publications from 2005, information was provided about additional compounds from the AAS group used in food supplements, e.g., stanazolol, metadienone, and nandrolone. Veterinary steroids such as boldenone and trenbolone have also been shown to be present. A previous study identified methylstenbolone (*Figure 1*) in dietary supplements, which is a new compound from the AAS group.[239,240]

Some previous studies contain information on food supplements containing AAS, which are available through online and offline purchase.

Studies on AAS showed a multidirectional mechanism of pharmacological activity, including the effects of androgen receptor activity, which correlate with the effect on intracellular signal transduction (from the cell membrane to the cell nucleus) as well as the regulation of transcription. The anabolic effect of these compounds is associated with the stimulation of protein biosynthesis, which causes retention of more nitrogen in the human body, resulting in the gain of muscle mass and overall body weight increase as well as enhancement of erythropoiesis.[241,242] Their androgenic effect includes the development of male sexual characteristics, e.g., facial hair. AAS have been used in the pharmacotherapy of impotency, hypogonadism, and delayed puberty in men. Because of its anabolic effects, the use of these compounds from the AAS group is recommended in cases of aplastic anemia, extensive burns, and frostbites, and in post-traumatic conditions with long-term immobilization, excessive emaciation, and the destruction of the human body in the course of cancer.[241,242]

Studies have shown that the use of AAS is associated with many side effects. Various compounds from the AAS group were identified in dietary supplements, which are intended for oral application at doses exceeding the recommended therapeutic doses. It has been shown that the use of food supplements containing various compounds from the AAS group and in uncontrolled doses increases the risk of side effects and enhances adverse reactions in the body. The most common side effects associated with the use of AAS include liver damage or disorder (congestive jaundice), gynecomastia, acne, change in voice tone, changes in body water and sodium levels, hypertension, increased hematocrit, and blood viscosity that are predisposed to blood clots, embolism, myocardial infarction, and stroke. Prolonged use of AAS causes changes in behavior and fluctuations in emotions, namely irritability, aggression, subdepressive or submaniac states, and increased and difficult to control sex drive. It has also been proven that AAS can inhibit spermatogenesis and steroidogenesis and con-

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tribute to testicular atrophy and prostate hyperplasia, and it may affect reproduction and fetus masculinization.[241,242] In previous scientific studies cases of hepatotoxicity and kidney damage have been reported in men using dietary supplements containing AAS.[243]

DHEA (prasterone) is a compound from the group of sex hormones and functions as a precursor of testosterone and estrogen. In the human body, DHEA is produced by the adrenal cortex, which produces androstedione and testosterone. Therapeutic agents containing DHEA are used to relieve the symptoms of andropause in men. DHEA remains on the WADA Prohibited List due to the significant risk of disturbances in the endocrine balance of the human body, i.e., the imbalance in the amounts of androgens/estrogens.[13] Many sports enthusiasts view DHEA as an alternative to AAS. Most studies have shown that DHEA has no significant effect on testosterone levels in men.[244] In a few studies such as those performed by Liu et al. an increase in testosterone in men was found during DHEA supplementation.[245]

Selective Androgen Receptor Modulators

Selective androgen receptor modulators (SARMs) are a new group of compounds with therapeutic potential. In 1998, bicalutamide was introduced, the first drug from the SARM group – the androgen receptor antagonist.[246,247]

Androgen receptor agonists are currently undergoing intensive research. Presently, they have been shown to have the ability to stimulate muscle growth, allow bone reconstruction, and reduce body fat. In terms of chemical structure, SARMs are a heterogeneous group of compounds. SARMs belong to the derivatives of arylpropionamide, bicyclic hydantoin, quinoline, and tetrahydroquinoline. SARMs are nonsteroidal compounds with anabolic activity. The compounds from this group of derivatives show selective action on organs such as muscles, bones, and the prostate. Unlike AAS, SARMs do not affect the activity of the aromatase and α-reductase enzymes. This makes SARMs exhibit limited androgenic and estrogenic effect; they do not cause side effects that are typical for AAS. SARMs do not cause hirsutism and virilization in women. In the future, SARMs may prove to be promising agents in the treatment of hypogonadism, osteoporosis, prostate hypertrophy, and muscle loss (cachexia) in the course of cancer or sarcopenia.[246,247]

On the basis of the conducted research, it was found that no structure classified as an SARM agonist was approved for use in pharmacotherapy. Most SARMs agonists have been tested *in vitro* or in animal models.[246,247] Few of them, marked with symbols S-4 and S-22, were included in clinical trials. Clinical research on the S-4 compound was suspended due to adverse effects – blurred vision. Compound S-22 (Enobosarm/Ostarine) was studied in a group of healthy people, i.e., elderly men and women in the postmenopausal period. Studies have shown that compound S-22 reduces total cholesterol, increases tissue sensitivity to insulin, and improves the physical functions of the human body. The preparations used in the therapy were well tolerated, and the most common side effects were nonspecific symptoms such as back pain, headache, pain in extremities, or flu-like symptoms. The study conducted showed a decrease in testosterone levels in men and a decrease in follicle-stimulating hormone (FSH, follitropin) and luteinizing hormone (LH, lutropin) in women. Hematological changes were found in isolated cases. Compound S-22 is in phase III clinical trials in a group of patients with lung cancer. Further research is required regarding the safety of SARMs and the possible consequences that may occur with their chronic use.[247] In 2008, SARMs were placed on WADA Prohibited List of doping agents because of their anabolic effect.[13] S-4 (Andarine) is a compound from the SARMs group, which is one of the first to be identified in the dietary supplements sold online.[248] The first case of SARMs detection during doping control was encountered in 2010. In other scientific studies, cases of the use of SARMs as a prohibited supplement in sports have also been reported. The fact that SARMs is the current problem of doping in sport is confirmed by the results of anti-doping controls in 2019.[249] The products available for sale online and containing various compounds from the SARMs group include Andarine (*Figure 1*), Ostarine, Ligandrol, and Testolone. Some products contained more than one of the SARM compounds. In the analyzed food supplements, other active compounds, not classified as SARMs, were also identified, e.g., ibutamoren, which stimulates human growth hormone (GH).[250]

Conclusions

Food supplements categorized as 'pre-workout, fat burners, brain boosters, and hormone boosters' have the highest risk of containing substances harmful for

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the human body, which pose a threat to the health and even life of the consumer. Adverse effects may result not only from the presence of a given component, but also from their interaction with other ingredients of dietary supplements, classic foods, or medical products. There are dietary supplements available in the market that contain a dangerous and unhealthy combination of active ingredients in one serving, each of which has a different mechanism of biological activity (gripping point) but directed at stimulating the human body. An example is the combination of substances such as DMAA, synephrine, higenamine, yohimbine, and huperzine in one product. In other cases, the use of food supplements may be the cause of sanctions for doping (accidental/unintended) in professional sports.

In other cases, dietary supplements can contain plant ingredients that are not traditionally used by humans and often work as a 'camouflage' for stimulating substances. Such examples include *Aele marmelosa* (aegeline), *Acacia rigidula*, *Dendrobium* sp., *Geranium* sp., *Eria jarensis*, *Kigelia africana*, *Pelargonium graveolens*, and *Senegalia berlandieri*. New plant extracts, however, are known in traditional medicine in Asia, Australia, or Africa, e.g., *Evodia rutaecarpa*, *Eurycoma longifolia*, *Piper methysticum*, *Sceletium tortuosum*, *Huperzia serrata* (huperzine), *Nelumbo nucifera* (higenamine), *Griffonia simplicifolia* (5-hydroxytryptophan), and *Mucuna* sp. (L-DOPA).

Substances included in food supplements show different mechanisms of biological activity in the human body. Dietary supplements called *testosterone boosters* have the ability to modulate the functioning of the endocrine system. Fat burners and pre-workout dietary supplements have been proved to affect the peripheral nervous system and the CNS, and they can indirectly disturb the functioning of the circulatory system, the activity of the liver and the kidneys, and thermoregulation. Dietary supplements often contain different biologically active substances that can regulate the activity of neurotransmitters and the following systems: the adrenergic system, the dopaminergic system, the serotonergic system, the cholinergic system, the GABAergic system, the glutamatergic system (NMDA), and the adenosine system.

Author Contribution Statement

K. J. and B. M. designed and directed the project. K. J. and J. L. collected the literature data. J. L. prepared the graphical part of MS. All authors discussed the results and contributed to the final manuscript.

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